Jiuyi (Joey) Xu

Updated November, 3rd, 2026

Colorado School of Mines
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EDUCATION

Ph.D. Robotics, Colorado School of Mines, Golden, CO — 08/2024 - Present
- Research Interest: **Vision-Language-Action (VLA) Models, Efficient AI**
- Advisor: Dr. Yangming Shi

M.S. Computer Science, University of Southern California, Los Angeles, CA — 08/2022 - 05/2024

B.E. Software Engineering, Dalian University of Technology, Dalian, China — 09/2017 - 06/2021

RESEARCH EXPERIENCE

Yangming Research Group, Colorado School of Mines — **Golden, CO**
Graduate Research Assistant, Advisor: Dr. Yangming Shi — 08/2024 - Present
- Improve the efficiency of embodied foundation models, including vision-language-action (VLA) and world-action models (WAMs), in a training-free manner.
- Accelerate inference through token compression, post-training quantization, and efficient sampling without compromising closed-loop task performance.
- Apply computer vision and generative modeling to construction.

Geospatial Terrain Research Group, USC Institute for Creative Technologies — **Los Angeles, CA**
Research Assistant (Intern), Advisor: Dr. Meida Chen — 06/2023 - 06/2024
- Developed open-vocabulary frameworks for automated 2D and 3D data annotation and segmentation.
- Reconstructed large-scale 3D terrain with Gaussian splatting and incrementally updated scenes from new imagery.

WORK EXPERIENCE

Prime Robotics — **Denver, CO**
Robotics Intern, Manager: Ryan Austin — 05/2026 - 08/2026
- Developed and maintained core components of a real-time multi-agent path-planning system coordinating 10+ robots in production.
- Built LLM-driven robot agents via the Model Context Protocol (MCP), enabling robots to reason about and execute tasks from their own embodied perspective.

PUBLICATIONS

- **Xu, J.**, Hu, X., Chen, M., Wang, S., Ye, Y., & Shi, Y. (2026). *Quantize by Doing: Task-Guided Mixed-Precision Quantization of Vision-Language-Action Models*. ArXiv.org. https://arxiv.org/abs/2610.38661
- **Xu, J.**, Chen, M., Ye, Y., & Shi, Y. (2026). *CuraEval: Evaluating Demonstration Curation on Mixed-Skill Robot Data*. "Oops, I Erred" Workshop at the 2026 Conference on Robot Learning (CoRL). https://arxiv.org/abs/2610.38221
- **Xu, J.**, Hu, X., Chen, M., Gao, P., Ye, Y., & Shi, Y. (2026). *World Models Dream of Success: Diagnosing and Repairing Failure Insensitivity in Robot World Models*. ArXiv.org. https://arxiv.org/abs/2610.09134
- **Xu, J.**, Guo, J., Chen, M., Du, J., & Shi, Y. (2026). *Predict Before You Deploy: Offline Prediction of Quantization-Induced Task Degradation for World Action Models*. ArXiv.org. https://arxiv.org/abs/2609.19441
- **Xu, J.**, Jin, Q., Chen, M., Wang, S., Sui, Y., & Shi, Y. (2026). *VLAQuantBench: Closed-Loop Evaluation of Post-Training Quantization for Vision-Language-Action Models*. ArXiv.org. https://arxiv.org/abs/2609.25376

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- **Xu, J.**, Qian, C., Tu, Z., Sui, Y., & Shi, Y. (2026). *FastVLA: Task-Conditioned and Temporally Adaptive Visual Token Compression for Efficient Vision-Language-Action Models*. 2027 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV). https://arxiv.org/abs/2610.22146
- **Xu, J.**, Chen, M., & Shi, Y. (2026). *EgoGen: Generating Egocentric Images on Construction Sites Using Diffusion Model*. Construction Research Congress 2026, 63–72. https://doi.org/10.1061/9780784486962.007
- Vahidi, M., **Xu, J.**, Shi, Y., & Liu, K. (2026). *Domain Knowledge-Infused Prompting for Deep and Full-Sentence Information Extraction from Bridge Inspection Reports*. Journal of Computing in Civil Engineering.
- Yu, H., **Xu, J.**, Shi, Y., & Ye, Y. (2026). *Principal Variate Selection Approach for LSTM Building Load Forecasting*. Proceedings of SimBuild 2026: 12th National Conference of IBPSA-USA, 327–338. https://doi.org/10.26868/30680611.2026.1314
- **Xu, J.**, Kim, R., Ye, Y., & Shi, Y. (2026). *An Autoregressive-based Framework for Simulating Stochastic Occupant Behavior in Residential Buildings*. Building Simulation. Vol. 19. No. 2. Tsinghua University Press, 2026. https://link.springer.com/article/10.1007/s12273-026-1406-3
- **Xu, J.**, Jin, Q., Chen, M., Feng, A., Sui, Y., & Shi, Y. (2025). *LowDiff: Efficient Diffusion Sampling with Low-Resolution Condition*. ArXiv.org. https://arxiv.org/pdf/2509.15342
- **Xu, J.**, Chen, M., & Shi, Y. (2025). *GenVision: Enhancing Construction Safety Monitoring with Synthetic Image Generation*. 2025 Winter Simulation Conference (WSC). IEEE, 2025. https://ieeexplore.ieee.org/abstract/document/11338881
- Chen, M., Leal, L., Hu, Y., Liu, R., Xiong, B., Feng, A., **Xu, J.**, & Shi, Y. (2025). *IDU: Incremental Dynamic Update of Existing 3D Virtual Environments with New Imagery Data*. 2025 Interservice/Industry Training, Simulation and Education Conference (I/ITSEC). https://arxiv.org/abs/2508.17579
- Liu, Z., **Xu, J.**, Wun, C., Chen, M., Zou, Z., & Shi, Y. (2025). *Egocentric Camera-Based Method for Detecting Static Hazardous Objects on Construction Sites*. Automation in Construction 172 (2025): 106048. https://www.sciencedirect.com/science/article/pii/S0926580525000883
- **Xu, J.**, Chen, M., Feng, A., Yu, Z., & Shi, Y. (2024). *Open-Vocabulary High-Resolution 3D (OVHR3D) Data Segmentation and Annotation Framework*. 2024 Interservice/Industry Training, Simulation and Education Conference (I/ITSEC). https://arxiv.org/abs/2412.06268
- **Xu, J.**, Sanni, T., Liu, Z., Yang, Y., Lee, J., Song, W., & Shi, Y. (2024). *To Shelter or Not To Shelter: Exploring the Influence of Different Modalities in Virtual Reality on Individuals' Tornado Mitigation Behaviors*. ArXiv.org. https://arxiv.org/abs/2409.09205
- Chen, M., Lal, D., Yu, Z., **Xu, J**., Feng, A., You, S., Nurunnabi, A., & Shi, Y. (2024). *Large-Scale 3D Terrain Reconstruction Using 3D Gaussian Splatting for Visualization and Simulation*. The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, XLVIII-2-2024, 49–54. https://doi.org/10.5194/isprs-archives-xlviii-2-2024-49-2024
- Rahman, M. A., **Xu, J.**, Liu, Z., Chen, M., Zhu, R., Liu, Y., Shi, Y., & Feng, A. (2024). *An Open-Vocabulary Framework for Efficient 2D and 3D Visual Data Annotation in Indoor and Outdoor Built Environments*. 2024 ASCE International Conference on Computing in Civil Engineering (i3CE).
- Liu, Z., **Xu, J.**, Suen, W. K., Chen, M., Zou, Z., Feng, A., & Shi, Y. (2024). *Detecting without Training: An Open-Vocabulary Object Detection Method for Identifying Hazardous Objects on Construction Sites*. 2024 ASCE International Conference on Computing in Civil Engineering (i3CE).

CONFERENCE PRESENTATIONS

- "EgoGen: Generating Egocentric Images on Construction Sites Using Diffusion Model", *CRC 2026*, San Antonio Marriott Rivercenter on River Walk, San Antonio, TX, March 18th – 21st, 2026
- "GenVision: Enhancing Construction Safety Monitoring with Synthetic Image Generation", *WSC 2025*, Sheraton Grand Seattle, Seattle, WA, December 7th – 10th, 2025

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- "An Open-Vocabulary Framework for Efficient 2D–3D Visual Data Annotation in Indoor and Outdoor Built Environments", *i3CE 2024*, Carnegie Mellon University, Pittsburgh, PA, July 28th – 31st, 2024
- "A Hybrid 2D–3D Zero-shot Framework for Photogrammetric Data Segmentation", *Geo Week 2024*, Colorado Convention Center, Denver, CO, February 11th – 13th, 2024
- "USC-ICT Semantic Terrain Points Labeling System", *I/ITSEC 2023*, West Concourse of the Orange County Convention Center, Orlando, FL, November 27th – December 1st, 2023

PEER REVIEW EXPERIENCE

- Journals: Journal of Computing in Civil Engineering, IEEE Transactions on Neural Networks and Learning Systems, International Journal of Data Science and Analytics
- Conferences: NeurIPS 2026, AISTATS 2026, AISTATS 2027, ICASSP 2027

AWARDS

- NVIDIA Academic Grant (2026)
- American Society of Civil Engineers (ASCE) Outstanding Reviewer (2025)
- American Society of Civil Engineers (ASCE) Best Data Award (2025)